Dense Flow from Adaptive Correspondence

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Abstract

Reliable optical flow requires combining evidence across pixels without mixing different motions. Modern networks strengthen matching and the use of context, but generally retain fixed-grid correspondence units and link their matching evidence to pixel motion indirectly. We introduce *Dense flow from Adaptive CorrespondencE* (DACE), which learns the pixel support of correspondence units jointly with motion estimation. Shared soft pixel-to-unit assignments pool matching features, define unit-flow targets, and distribute predicted flow to pixels. Assignments are resolved at image resolution, while matching and refinement operate over fewer units. Under the standard zero-shot protocol, DACE achieves the lowest average flow error on Sintel Final and KITTI among the compared methods. We also find that a small fraction of pixels with occluded or out-of-frame targets accounts for roughly half of the average flow error in a fine-grid baseline. DACE reduces error in these regions and near motion boundaries. Controlled studies support adaptive units as a way to use context while preserving distinct motions.

1 Introduction

Optical flow infers dense motion from observations that constrain some pixels more reliably than others. Lucas–Kanade [2] pools local image constraints to resolve ambiguous displacement, while Horn–Schunck [3] couples motion estimates through spatial smoothness. Black–Anandan [4] emphasizes the opposing risk: integration across motion boundaries introduces competing constraints. This longstanding tension persists in learned flow estimation. Edges, weak texture, and missing correspondences require information beyond the individual pixel, yet nearby pixels need not share motion (Fig. 1). Reliable flow must therefore resolve ambiguity through spatial context without blurring motion discontinuities.

Modern flow networks have substantially improved motion estimation through richer matching evidence and more effective use of context. A common design takes fixed grid cells as the correspondence units and learns the features and prediction mechanisms operating on them. Within this framework, cost-volume and global-matching approaches improve correspondence search and estimation [5–9]. Context aggregation, pretrained representations, and learned priors strengthen inference under uncertain observations [10–14]. Multi-scale processing and higher-resolution feature indexing improve spatial detail [1, 15, 16]. Together, these advances strengthen matching evidence and motion inference around fixed-grid correspondence units.

However, these advances largely leave the longstanding tension between sharing evidence and separating motions to feature learning and flow upsampling. In these approaches, evidence aggregation and flow upsampling establish separate associations between each unit and the pixels it represents. Coarse units can span motion boundaries; fine indexing improves localization, but weak or missing matches still require evidence beyond the individual pixel. Explicit motion-feature aggregation helps recover missing information, yet operates on evidence already formed on fixed-grid units [10]. Whether a unit's matching evidence is compatible with the pixel motion it represents thus remains an indirect outcome of learning, especially where observations are weak and neighboring motions conflict.

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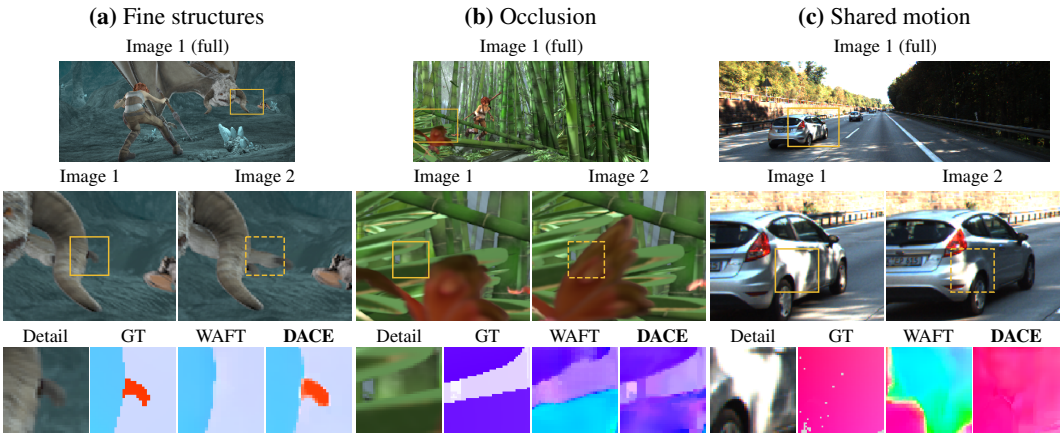

Figure 1: **Optical flow must combine compatible evidence while separating different motions.** (a) A fine structure moves differently from its surroundings. (b) A wing hides a background patch in Image 2, removing its visible match. (c) Different appearances across the car body share coherent motion. Gold boxes locate aligned crops; bottom rows compare ground truth (GT) with zero-shot WAFT [1] and DACE predictions. Colors share a scale per example; gray marks unlabeled GT.

We address this longstanding tension by learning the correspondence unit itself, explicitly coupling its matching evidence with the pixel motion it represents. We introduce *Dense flow from Adaptive CorrespondencE* (DACE), which learns which pixels constitute each correspondence unit (Fig. 2). Classical results on local grouping also show the value of adapting spatial support for motion estimation [17]. DACE learns grouping and matching evidence jointly from flow supervision, forming units to gather compatible evidence and represent locally coherent motion.

DACE realizes this coupling through shared soft pixel-to-unit assignments. A compact per-pixel parameterization predicts these assignments at image resolution, while matching and iterative refinement operate over fewer adaptive units. The same assignments pool features for matching, define unit-flow targets, and distribute predicted unit flow to pixels. During flow training, the assignments, matching features, and unit predictions are learned jointly for dense flow estimation. Together, these operations help uncertain pixels draw on compatible evidence while maintaining the separation between different motions.

Under the standard zero-shot protocol, DACE achieves the lowest average flow error on Sintel Final and KITTI among the compared published methods. With fewer refinement steps, it also outperforms a fine-grid baseline [1] while requiring less computation. Our regional analysis identifies an error concentration obscured by overall scores: occluded or out-of-frame pixels occupy only 6.4–13.9% of valid pixels yet contribute 42.3–52.2% of this baseline's average flow error across the evaluated datasets. DACE consistently achieves larger absolute error reductions in these regions and near motion boundaries than in the overall average.

Adaptive units improve both the use of local context and the preservation of motion detail. At the same unit count, DACE outperforms fixed-grid controls and benefits more from local context. Its learned pixel-to-unit associations also provide a more faithful representation of dense motion than fixed-grid units. These complementary gains support adaptive correspondence units as a formulation for addressing the longstanding tension between evidence sharing and motion separation.

Our main contributions are:

- We introduce adaptive correspondence units to address the longstanding tension between evidence sharing and motion separation, explicitly coupling matching evidence with represented pixel motion.
- We quantify how error in a fine-grid baseline concentrates on a small subset of pixels with occluded or out-of-frame targets, a pattern obscured by overall scores.
- We achieve the lowest zero-shot average flow error on Sintel Final and KITTI among the compared methods, with consistent gains without direct matches and near motion boundaries. Controlled studies support the formulation through better context use and motion preservation.

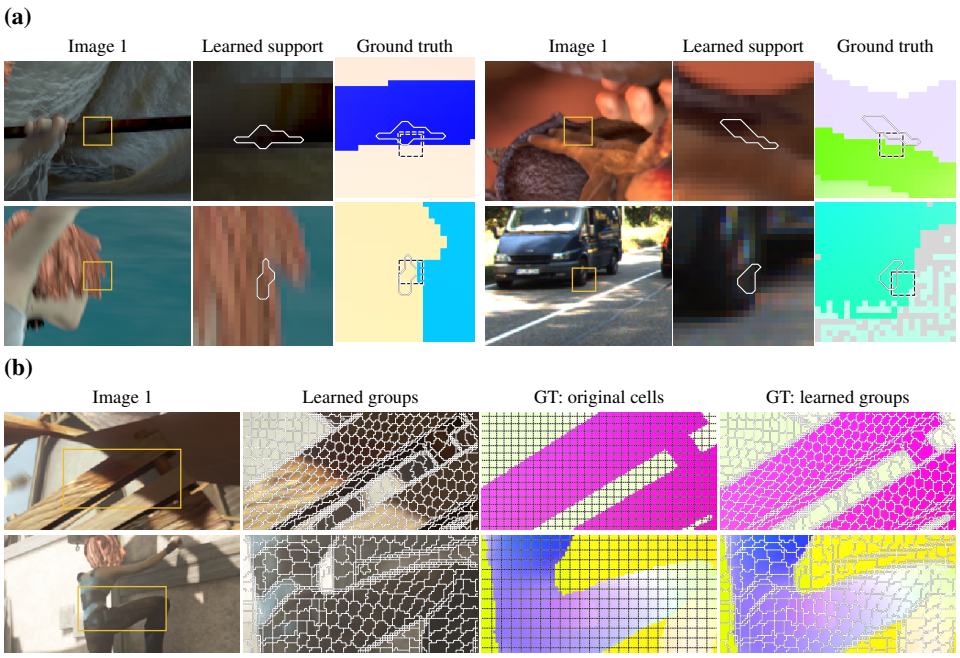

Figure 2: **DACE learns which pixels each correspondence unit represents.** (a) Individual supports follow fine structures and object boundaries. White contours show learned support; black dashes mark original lattice cells. (b) Neighboring groups follow slat edges and a bent body outline, separating foreground and background motion. The two right columns show identical ground-truth flow (GT); only the partition changes. All groups in each crop are shown. Gold boxes locate aligned crops. Contours show strongest assignment; the model retains soft assignments. All examples use the zero-shot model; gray GT pixels are unlabeled.

2 Related Work

Spatial support for motion estimation. Classical optical flow couples local measurements to overcome ambiguous or missing evidence. Lucas–Kanade pools constraints within a neighborhood, Horn–Schunck couples neighboring estimates through smoothness, and Black–Anandan uses robust penalties to accommodate competing motions [2–4]. Ren makes spatial support explicit through contour-derived soft affinities for local affine estimation, with motion-based boundary ownership refining these affinities [17]. DACE jointly learns which pixels form each correspondence unit, the matching evidence they provide, and the motion the unit represents within an end-to-end flow estimator.

Matching evidence and context. Modern two-frame networks strengthen matching evidence through learned pyramids and cost volumes [5, 6], global matching and cost-memory encoding [7, 8], and pretrained representations [12, 13]. Improved recurrent estimation and learned motion priors [9, 14] support estimation under uncertain observations. Source-image affinities aggregate motion features [10], while feature similarity guides flow propagation [7]. Multi-scale processing, learned flow upsampling, and high-resolution indexing improve spatial detail [1, 6, 15, 16]. These approaches strengthen matching and prediction around fixed-grid units, while feature aggregation and learned flow upsampling define their pixel associations separately. Although optimized end to end, they rely on the learned pipeline to make each unit's matching evidence compatible with the pixel motion it represents. DACE learns a shared pixel-to-unit association that explicitly couples these roles through feature pooling, unit-flow supervision, and dense prediction.

Learned superpixels. Superpixel methods focus on partitioning images into compact regions that follow image boundaries [18, 19]. Learned variants make these partitions differentiable and adaptable to downstream tasks [20, 21]. DACE learns groups as correspondence units for flow prediction. The same assignments pool matching features, define unit-flow targets, and distribute predicted flow to

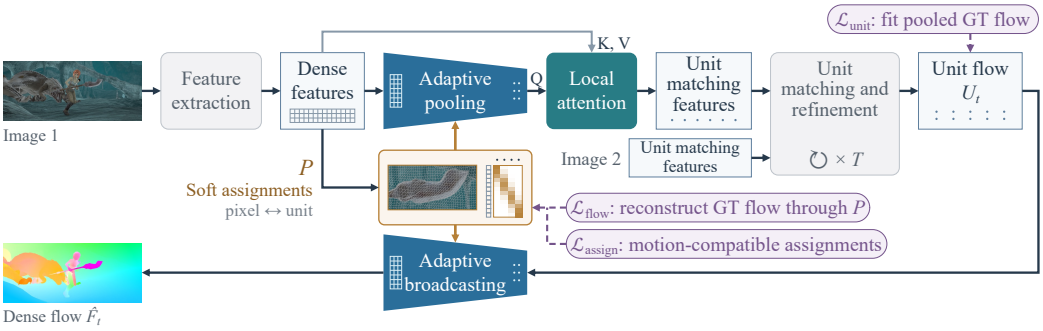

Figure 3: **DACE couples matching evidence and dense flow through learned pixel-to-unit assignments.** Soft assignments P, predicted from dense image features, define adaptive units: pooling through P forms each unit's matching features, and broadcasting through P returns the predicted unit flow U_t to pixels as the dense flow $\widehat{F}_t$. Matching and refinement operate over the coarser unit lattice (Image 2's units use shared weights), and flow supervision trains both the units and the assignments.

pixels, so grouping is learned jointly with matching to support both evidence aggregation and motion representation.

3 Method

Overview. DACE learns adaptive correspondence units that couple matching evidence to the pixel motion they represent, using context to resolve ambiguity while preserving motion boundaries. Given two images, we estimate dense flow from Image 1 to Image 2 by matching adaptive units and distributing their predicted flow to pixels (Figure 3). The same learned pixel-to-unit assignments pool dense features, define unit-flow targets, and distribute unit predictions to pixels. Assignments are resolved at image resolution, while matching and refinement operate over a coarser lattice of units. Flow supervision learns the assignments and shared image features jointly, shaping each unit to gather useful evidence and represent pixel motion.

3.1 Forming adaptive correspondence units

The same learned assignments determine which pixel features are pooled into each unit and how its predicted flow returns to pixels.

A shared pixel-to-unit association. Each unit has an anchor q on a regular lattice. The anchors fix the unit count and indexing; each unit's pixel support is learned. For one image with N pixels and M units, we represent this association by a soft assignment matrix $P \in \mathbb{R}_{+}^{N \times M}$ with weights P_{xq}. A row specifies pixel x's assignment weights to nearby units, and a column defines one unit's pixel support. Each pixel distributes its weight over the 3×3 anchor neighborhood centered on its own cell, $\mathcal{N}(x)$, with $\sum_{q \in \mathcal{N}(x)} P_{xq} = 1$; entries outside this neighborhood are zero. We omit boundary padding from this notation.

For a dense value map Z, pooling forms a weighted mean for each unit. For unit values U, broadcasting combines the unit values at each pixel. Both follow from the same soft assignment matrix:

$$\bar{Z} = D_\epsilon^{-1} P^\top Z, \quad \widehat{Z} = PU, \quad D_\epsilon = \operatorname{diag}(P^\top \mathbf{1}) + \epsilon I. \tag{1}$$

The diagonal matrix D_ϵ normalizes each pooled sum by the unit's total assignment weight, with a small $\epsilon > 0$ preventing division by zero. The row weights specify how unit values are shared with pixels. Applying this relation to features, flow targets, and predicted flow ties correspondence evidence to the pixels that the unit represents.

Predicting pixel-to-unit assignments. The assignment head learns which nearby units should represent each pixel, and how strongly. From the dense image features, it predicts a two-dimensional Gaussian over the candidate anchor positions for every pixel. The Gaussian's center shifts weight toward particular units; its spread and orientation control how that weight is shared among them.

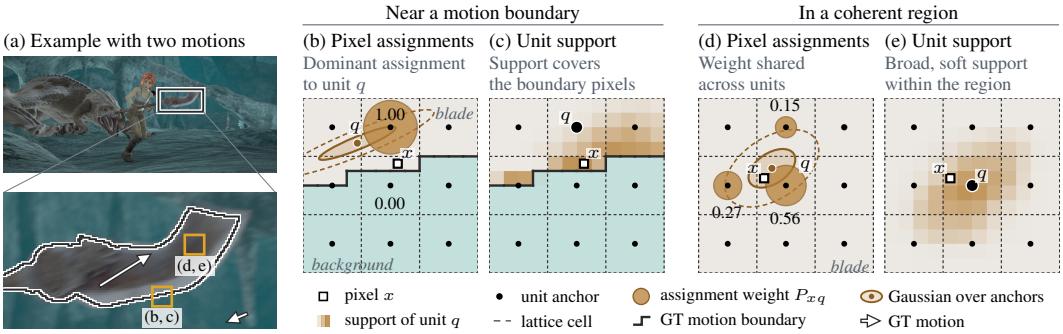

Figure 4: **Learned assignments adapt units to motion structure.** (a) The moving blade from Image 1 in Figure 3; gold boxes mark the two pixels x examined in (b–e). (b, d) Assignment weights of x over nearby anchors, a row of P. (c, e) Support of its most strongly associated unit q, a column of P. Near the motion boundary, x favors a neighboring blade unit over its own cell's background unit, so pixels of one cell contribute to different units; the support of q follows the boundary. Inside the blade, several units share its weight, giving broad, soft support.

For pixel x, we take the anchor of its own cell as the local origin. Each candidate anchor q has a fixed offset $\delta_{xq} \in \{-1, 0, 1\}^2$ from this origin. A compact head predicts five parameters: the two components of the mean offset $\mu_x \in \mathbb{R}^2$ and three entries of a lower-triangular factor L_x with positive diagonal. The latter defines the inverse covariance $G_x = L_x L_x^\top$.

Evaluating the Gaussian gives scores a_{xq} at the candidate anchors, which are normalized into assignment weights:

$$a_{xq} = -\tfrac{1}{2}(\delta_{xq} - \mu_x)^\top G_x (\delta_{xq} - \mu_x), \qquad P_{xq} = \frac{\exp a_{xq}}{\sum_{r \in \mathcal{N}(x)} \exp a_{xr}}. \tag{2}$$

Each pixel predicts its own Gaussian. A unit's support is the corresponding column of P, collecting its assignment weights across all pixels. This support can therefore take an irregular shape and extend across nominal cell borders.

Figure 4 connects these two views: a pixel's assignments over anchors (b, d) and the support of its most strongly associated unit (c, e). Near the illustrated motion boundary, a pixel favors a neighboring unit whose support follows the boundary; inside the coherent region, several units share its weight. Pixels within one lattice cell can therefore contribute to different units and receive different combinations of unit flows. These assignments provide pixel-level control over evidence aggregation and flow sharing near fine structures and motion boundaries.

3.2 Matching adaptive units

We perform matching and flow refinement in unit space: each adaptive unit provides matching features and a flow estimate for its learned pixel support.

Constructing unit matching features. Within each image, pooling summarizes dense features over each unit's learned pixel support. The pooled feature serves as a query for local attention over nearby dense features in the same image. This enriches the unit's matching evidence with local context, while its assignments determine how the predicted unit flow is shared among pixels. We apply this construction separately to both images with shared network weights.

Matching units and predicting pixel flow. We estimate unit flow from the two adaptive-unit feature maps using the iterative matching and refinement framework of WAFT [1]. Each image's assignments are predicted once and reused across updates.

Let U_t collect Image 1's predicted unit flows at update t, and let P denote its assignment matrix. Broadcasting through P gives the dense flow:

$$\widehat{F}_t = PU_t. \tag{3}$$

Each pixel receives the combination of unit flows specified by its row of P. Matching and refinement therefore operate in the coarser unit space, while the learned assignments link the evidence forming each unit to the pixels receiving its predicted flow. This lets pixels with weak or missing direct matches draw on evidence gathered for their assigned units, while pixel-level assignments control flow sharing near motion boundaries.

3.3 Joint learning of unit flow and pixel assignments

We train unit flow and pixel-to-unit assignments jointly, aligning the evidence each unit gathers with the pixel flow it represents. Three complementary losses supervise unit predictions, pixel assignments to compatible predictions, and the motion structure preserved by the grouping. We use ℓ for per-unit or per-pixel losses and $\mathcal{L}$ for their aggregated training objectives. Below, F^* stacks Image 1's ground-truth pixel flows. We omit the refinement-step index in the prediction losses until the joint objective.

Supervising unit flow. Each unit should predict the flow of its current pixel support. Pooling ground-truth flow through P gives the unit targets U^*. For training, each predicted unit flow U_q is accompanied by uncertainty parameters, defining likelihoods centered on its two flow components, following SEA-RAFT [9]. We denote these likelihoods collectively by p_q and use ℓ_{NLL} for negative log-likelihood averaged over the two flow components. The stop-gradient operator sg holds the pooled target fixed during backpropagation:

$$U^* = \mathrm{sg}\big(D_\epsilon^{-1} P^\top F^*\big)\,, \qquad \ell_{\mathrm{unit}}(q) = \ell_{\mathrm{NLL}}\big(p_q, U_q^*\big)\,. \tag{4}$$

Detaching the target prevents grouping from reducing this loss by moving it toward the prediction. Gradients still pass through feature aggregation and matching, teaching the support to supply useful correspondence evidence.

Learning pixel assignments from unit predictions. We train pixel-to-unit assignments to favor neighboring unit predictions compatible with each pixel's ground-truth flow. Even an accurate prediction of a unit's average flow can disagree with individual pixels near a motion boundary. For pixel x, we combine neighboring unit likelihoods according to its learned assignment weights to form a pixel likelihood $\widetilde{p}_x$. Mixing is performed separately for each flow component. We hold the unit flow and uncertainty predictions fixed during backpropagation and supervise the assignments with the pixel's ground-truth flow:

$$\ell_{\mathrm{assign}}(x) = \ell_{\mathrm{NLL}}(\widetilde{p}_x, F_x^*)\,. \tag{5}$$

Without detachment, this loss could update both the assignments and unit predictions. We deliberately separate their direct supervision: the unit loss trains predictions against pooled flow targets, while the assignment loss trains compatible pixel-to-unit weights and shared image features. Our assignment stop-gradient ablation supports this choice.

Preserving motion structure in the grouping. We train the grouping to preserve motion differences between pixels through pooling and broadcasting. Applying both operations to ground-truth flow provides a direct training signal independent of the current unit predictions. Following SSN's reconstruction principle [20], we compare the reconstructed flow F_P^* with the original:

$$F_P^* = P\big(D_\epsilon^{-1} P^\top F^*\big)\,, \qquad \ell_{\mathrm{flow}}(x) = \ell_{\mathrm{rec}}((F_P^*)_x, F_x^*)\,, \tag{6}$$

where ℓ_{rec} is a Charbonnier penalty. Gradients pass through both uses of P. This penalizes motion differences lost through pooling and broadcasting, teaching the support to preserve motion boundaries.

Joint objective. Let $\mathcal{L}_{\mathrm{unit}}$, $\mathcal{L}_{\mathrm{assign}}$, and $\mathcal{L}_{\mathrm{flow}}$ denote the respective losses aggregated over units or pixels. We supervise unit flow and assignments at every refinement step with weights γ^{T-t}, where T is the number of updates, and add the flow-reconstruction loss once with weight λ_{flow}:

$$\mathcal{L} = \sum_{t=1}^{T} \gamma^{T-t}(\mathcal{L}_{\mathrm{unit},t} + \mathcal{L}_{\mathrm{assign},t}) + \lambda_{\mathrm{flow}}\mathcal{L}_{\mathrm{flow}}. \tag{7}$$

All three objectives shape the shared trainable features and assignments jointly. Together, they train units to gather useful matching evidence and estimate flow, while learning pixel assignments that preserve distinct motions.

4 Experiments

Our experiments demonstrate the value of explicitly coupling a correspondence unit's matching evidence with the pixel motion it represents. We first evaluate zero-shot performance, then identify where the fine-grid baseline's errors concentrate and examine DACE's improvements in these regions and near motion boundaries. Fixed-grid comparisons and ablations examine how the unit formulation and its design choices contribute to these gains. We then evaluate the trade-off between accuracy and computation.

4.1 Experimental setup

Training and evaluation. All comparisons use two input frames. We train DACE on FlyingChairs followed by FlyingThings3D and evaluate zero-shot on the Sintel and KITTI training splits, without target-dataset flow training. The FlyingThings3D test split (Things) provides an in-distribution comparison. Things and Sintel each have Clean and Final renderings, abbreviated C and F in tables. The main experiments use a DINOv3 ViT-S backbone; Appendix A repeats the zero-shot comparison with Depth Anything V2 [22].

Reference systems. Our implementation uses the iterative matching and refinement framework of WAFT [1], with five updates during training and, unless stated otherwise, inference. These systems extract features on an $(H/2) \times (W/2)$ grid for an $H \times W$ image, abbreviated $H/2$. Released WAFT estimates flow on this fixed grid, whereas DACE uses adaptive units anchored on an $H/4$ lattice, with a quarter as many correspondence units. The H/4 study compares adaptive and fixed-grid units using the same backbone, dense-feature resolution, unit count, iterative refinement framework, and number of updates. Both formulations are evaluated with and without local attention over the dense features. We use the released WAFT checkpoint and train the H/4 controls with its code and loss. Locally trained models follow its training data and optimization recipe; the broad comparison retains published results.

Metrics. Endpoint error (EPE) is the Euclidean distance between predicted and ground-truth flow vectors, in pixels, averaged within each image and then over images. KITTI Fl is the percentage of valid pixels whose error exceeds both 3 pixels and 5% of the ground-truth motion. Regional EPE pools errors over valid pixels in the specified region. We evaluate visible targets, occluded or out-of-frame targets, and neighborhoods of motion boundaries; the last can overlap either visibility region. On Things, the missing-target region includes only out-of-frame pixels. To assess accuracy near motion boundaries, we report EPE over valid pixels in 5×5 neighborhoods of locations whose ground-truth flow-change strength exceeds 2 pixels. Appendix B.2 gives the exact construction and additional thresholds; Appendix B defines the visibility masks. Computation is measured in multiply–accumulate operations (MACs) for one 540×960 image pair.

4.2 Zero-shot transfer

Dense flow ground truth is difficult to acquire in real scenes [23, 24]. Zero-shot evaluation therefore tests an important practical capability: accurate flow in new domains without target-domain flow supervision.

Adaptive correspondence transfers to unseen scenes. Under the standard zero-shot protocol, DACE achieves the lowest Sintel Final and KITTI EPE among the methods in Table 1. The gains span unseen rendered scenes and real driving imagery. The second-backbone comparison also lowers EPE relative to the fine-grid baseline on both Sintel renderings and KITTI (Appendix A). Together, these results show improved cross-domain flow prediction without target-specific fine-tuning, with benefits beyond a single backbone.

4.3 Missing matches and motion boundaries

Reliable flow requires evidence beyond the individual pixel when its own match is uncertain, together with precise separation of neighboring motions. Table 2 examines these demands by reporting EPE for occluded/out-of-frame pixels and for our defined neighborhoods of ground-truth motion boundaries.

Table 1: **DACE achieves the lowest Sintel Final and KITTI EPE among the compared methods.** Zero-shot results on the Sintel and KITTI training splits after training on FlyingChairs and FlyingThings3D. Bold: lowest error in each column. –: not reported.

Method	Sintel		KITTI		MACs
	Clean	Final	EPE	Fl (%)	(G)
FlowNet2 [25]	2.02	3.14	10.10	30.4	230
PWC-Net [5]	2.55	3.93	10.40	33.7	101
RAFT [6]	1.43	2.71	5.04	17.4	938
GMA [10]	1.30	2.74	4.69	17.1	1352
SKFlow [11]	1.22	2.46	4.27	15.5	751
DIP [26]	1.30	2.82	4.29	13.7	3068
CRAFT [27]	1.27	2.79	4.88	17.5	2274
GMFlowNet [28]	1.14	2.71	4.24	15.4	1094
FlowFormer [8]	1.01	2.40	4.09	14.7	1715
EMD-L [29]	0.88	2.55	4.12	13.5	1755
FlowFormer++ [30]	0.90	2.30	3.93	14.1	1713
MatchFlow(G) [12]	1.03	2.45	4.08	15.6	1669
AnyFlow [15]	1.10	2.52	3.76	12.4	–
RPKNet [31]	1.12	2.45	–	13.0	137
CCMR+ [32]	0.98	2.36	–	12.9	12653
SEA-RAFT(L) [9]	1.19	4.11	3.62	12.9	655
SAMFlow [13]	0.87	2.11	3.44	12.3	9717
FlowDiffuser [14]	**0.86**	2.19	3.61	11.8	2466
DPFlow [16]	1.02	2.26	3.37	**11.1**	414
WAFT [1]	1.28	2.56	3.49	12.9	732
DACE	1.03	**2.10**	**3.11**	11.9	796

Table 2: **DACE lowers EPE where no match is visible and near motion boundaries.** EPE is pooled over the pixels of each region. For (c), we define a region of valid pixels in 5×5 neighborhoods of ground-truth motion boundaries, using a 2-pixel local flow-change threshold (Appendix B.2). This region can overlap (a) and (b). Share: the region's percentage of valid pixels.

	(a) Visible		(b) Occluded/out-of-frame			(c) Near motion boundaries		
Dataset	WAFT	DACE	Share (%)	WAFT	DACE	Share (%)	WAFT	DACE
Things C	2.14	**1.37**	6.4	23.03	**16.61**	9.3	13.40	**9.24**
Things F	2.09	**1.40**	6.4	21.96	**15.89**	9.3	13.14	**9.48**
Sintel C	0.65	**0.49**	8.4	7.95	**6.77**	6.2	7.30	**6.63**
Sintel F	1.59	**1.28**	8.4	12.86	**10.93**	6.2	11.92	**10.45**
KITTI	**2.01**	2.05	13.9	13.65	**11.05**	13.6	9.97	**8.51**

A small pixel subset accounts for a large share of error. Occluded or out-of-frame pixels occupy only 6.4–13.9% of valid pixels, yet contribute 42.3–52.2% of the fine-grid baseline's standard EPE across the evaluated datasets (Appendix B). The overall score alone does not reveal this concentration. This breakdown identifies motion estimation without direct matches as a substantial remaining source of error even at fine correspondence resolution.

Recovering motion without visible matches. Occlusion and out-of-frame motion remove the target observation, so estimating the affected pixels' motion requires context. DACE lowers EPE in these regions in every evaluated dataset and rendering. On Things and Sintel, the absolute reductions are substantially larger than on visible pixels. On KITTI, visible-pixel error remains similar (2.05 versus 2.01), while missing-target error falls from 13.65 to 11.05. Thus, the overall gains include a substantial improvement precisely where direct matching evidence is unavailable.

Preserving motion boundaries. Boundary accuracy is particularly sensitive to which pixels a unit's motion represents. Within a region of nearly uniform flow, sharing motion among nearby pixels changes the result little; across a motion boundary, the same sharing can assign pixels the motion of the wrong surface. Evaluating EPE in the selected boundary neighborhoods therefore tests

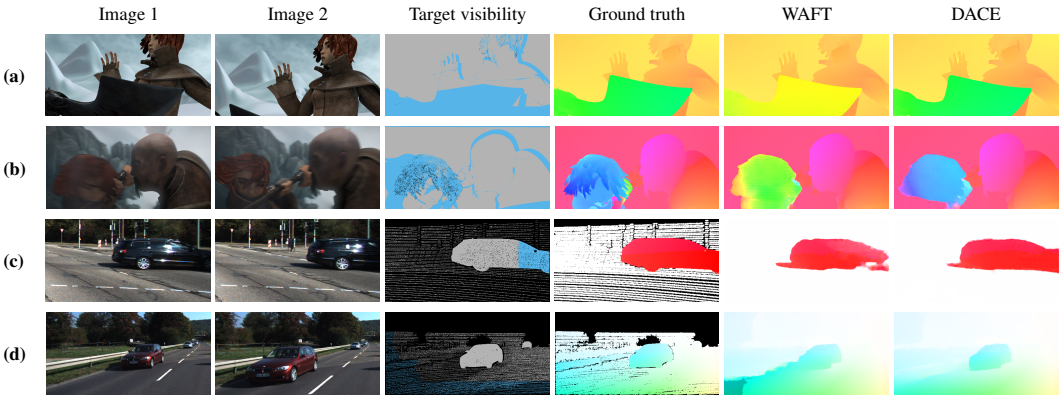

Figure 5: **DACE improves motion recovery and boundary precision.** Zero-shot examples show an occluded cape (a, Sintel Clean), blurred hair (b, Sintel Final), a vehicle leaving the image (c, KITTI), and an approaching vehicle (d, KITTI). All panels within a row share one crop; (c) retains the original right image boundary. Blue marks occluded/out-of-frame pixels, gray other valid pixels, and black excluded pixels. Flow colors match within each pair; sparse KITTI labels are enlarged only for display.

whether predicted motion is assigned to the right pixels near a discontinuity. DACE lowers EPE in these neighborhoods in every evaluated dataset and rendering, despite using a quarter as many correspondence units as the fine-grid baseline. The advantage persists at the additional boundary thresholds and against the H/4 control with local attention (Appendix B.2). Figure 4(b,c) illustrates this spatial adaptability: a boundary pixel favors a neighboring unit whose support follows the blade's motion boundary. Together, the regional results show stronger motion estimation without direct matches and more accurate separation of neighboring motions.

Qualitative comparison. Figure 5 illustrates improved motion recovery and boundary precision on Sintel and KITTI. DACE recovers the motion of the occluded cape and blurred hair, and better separates the vehicles' motion from the surrounding road. Appendix C adds complementary flow comparisons, learned-support maps and assignment-entropy views.

4.4 Adaptive versus fixed-grid correspondence

With the shared experimental basis above, we compare adaptive and fixed-grid correspondence at the same $H/4$ unit count. We examine overall accuracy, the benefit of local context, and how faithfully the units retain pixel motion.

Table 3: **Adaptive units improve flow accuracy and retain dense motion at the same unit count.** The systems share a backbone, feature resolution and training recipe. (a) Overall accuracy. (b) Occluded/out-of-frame EPE; attention gain is without minus with local attention. (c) Image-mean EPE after pooling ground-truth flow into $H/4$ units and applying each model's image-predicted readout, without retraining.

(a) Overall accuracy

		Things		Sintel		KITTI	
System	Attn.	C	F	C	F	EPE	Fl (%)
WAFT-H/4	–	2.93	2.72	1.08	2.49	3.96	13.3
WAFT-H/4	✓	2.73	2.62	1.07	2.57	3.75	12.6
DACE	–	2.56	2.48	1.12	2.14	3.54	13.0
DACE	✓	**2.39**	**2.38**	**1.03**	**2.10**	**3.11**	**11.9**

(c) Supplied-motion reconstruction EPE

WAFT-H/4	✓	0.50	0.53	0.21	0.39	1.52	—
DACE	✓	**0.19**	**0.21**	**0.09**	**0.19**	**0.39**	—

(b) Occluded/out-of-frame EPE

		Things		Sintel		KITTI
System	Attn.	C	F	C	F	EPE
WAFT-H/4	–	19.65	17.79	7.15	12.91	14.87
WAFT-H/4	✓	18.98	17.64	7.19	12.90	14.09
Attention gain		0.67	0.15	−0.04	0.01	0.78
DACE	–	17.49	16.55	6.89	11.40	12.96
DACE	✓	16.61	15.89	6.77	10.93	11.05
Attention gain		**0.88**	**0.66**	**0.12**	**0.47**	**1.91**

Adaptive correspondence outperforms the fixed-grid systems. DACE outperforms both fixed-grid controls on all six metrics in Table 3(a). Without local attention in either system, it still improves

five of six metrics. Adding context to the fixed grid therefore does not reproduce the adaptive system's performance.

Learned support and local context are complementary. Local attention reduces occluded/out-of-frame EPE more in DACE than in the fixed-grid system on every evaluated dataset and rendering (Table 3(b)). The larger gains in DACE support adaptive units as a useful basis for gathering evidence when direct matches are missing.

Adaptive units preserve the motion they represent. To test how faithfully the units represent pixel motion, we replace their predicted flow with pooled ground-truth flow. Keeping the image-predicted assignments and readout weights, we pool ground-truth flow into each model's $H/4$ units and map it back to pixels. Reconstruction EPE measures the motion detail lost through pooling and readout, separately from the accuracy of correspondence estimation. Appendix D gives the protocol.

DACE yields lower reconstruction EPE than the fixed-grid control with local attention on every evaluated dataset and rendering (Table 3(c)). On Sintel Final, EPE falls from 0.39 to 0.19; on KITTI, from 1.52 to 0.39. These results support learning which pixels each correspondence unit represents: adaptive support and its associated readout retain distinctions in dense motion more faithfully at the same coarse unit count.

4.5 Learning effective correspondence units

Table 4 examines how grouping supervision, contextual evidence, and assignment learning make the units effective for flow, changing one design choice at a time.

Table 4: **Each design choice improves accuracy on most or all metrics.** Each variant changes one choice from DACE. Prediction detachment blocks the assignment loss's gradients to unit flow and uncertainty; the pooled unit target remains detached in every variant.

					Things		Sintel		KITTI	
Variant	Flow recon.	Local attn.	Prediction detachment	Assignment scores	C	F	C	F	EPE	Fl (%)
DACE	✓	✓	✓	Gaussian	**2.39**	**2.38**	1.03	**2.10**	**3.11**	**11.9**
No flow reconstruction	–	✓	✓	Gaussian	2.68	2.61	1.05	2.17	3.14	12.4
No local attention	✓	–	✓	Gaussian	2.56	2.48	1.12	2.14	3.54	13.0
No prediction detachment	✓	✓	–	Gaussian	2.49	2.46	**1.00**	2.18	3.64	12.3
Direct logits	✓	✓	✓	Direct	2.58	2.55	1.02	2.24	3.16	12.1

Flow reconstruction teaches the grouping to preserve motion. Flow-reconstruction supervision improves all six metrics over the variant trained with only the unit and assignment losses. Those losses already train the grouping through the prediction task; reconstructing ground-truth flow through the same assignments supplies a complementary signal about motion differences lost in pooling and broadcasting. The result supports supervising the motion represented by the grouping itself.

Local attention strengthens the evidence available to each unit. Local attention improves all six metrics relative to the variant without it. Pooling summarizes the features within a unit's learned support; attention supplements them with nearby evidence, while the assignments continue to define the pixels receiving its flow. This ablation establishes the value of contextual evidence within DACE, complementing the comparison of its benefit across unit formulations in Section 4.4.

Separating direct supervision improves joint learning. Detaching unit flow and uncertainty in the assignment loss improves five of six metrics; Sintel Clean favors allowing these gradients. The pooled unit target remains detached in this comparison. These results support the chosen roles: unit predictions learn from pooled flow targets, while pixel assignments learn which predictions are compatible with their motion. Shared features and assignments are still trained jointly through the complete objective.

Gaussian scores encode a useful spatial prior. Gaussian assignment scores improve five of six metrics over directly predicted logits, with Sintel Clean favoring direct logits. The Gaussian in Eq. (2) links a pixel's candidate scores through the relative positions of nearby anchors: its center, spread, and orientation determine how weight is shared among them. This spatial constraint guides pixel-to-unit assignment while allowing each unit's support to take an irregular shape.

4.6 Accuracy and computation

Resolving pixel assignments at image resolution lets DACE preserve spatial detail while allocating matching and refinement to fewer units. Table 5 examines how this allocation affects accuracy and cost.

Table 5: **Fewer units leave room for richer processing.** Costs are GMACs for one 540×960 image pair; errors are EPE except KITTI Fl. †: profiled only, not trained.

			Cost (GMACs)			Things		Sintel		KITTI	
System	Resolution	Updates	Attention	Refinement	Total	C	F	C	F	EPE	Fl (%)
WAFT-64 [1]	$H/2$	5	–	425.8	732.4	3.54	3.44	1.28	2.56	3.49	12.9
WAFT-128 + attn.†	$H/2$	5	57.0	943.9	1313.2	–	–	–	–	–	–
DACE-64	$H/4$	5	20.6	**293.8**	**636.3**	2.55	2.48	**1.03**	2.24	3.52	12.5
DACE-128	$H/4$	4	20.6	363.1	706.1	2.44	2.43	**1.03**	2.15	3.18	12.2
DACE-128 (main)	$H/4$	5	20.6	453.9	796.8	**2.39**	**2.38**	**1.03**	**2.10**	**3.11**	**11.9**

Fewer correspondence units allow richer processing. Suffixes denote recurrent-state and refinement width: WAFT-64 is the released reference, DACE-128 is our main model, and DACE-64 uses a narrower state. Widening the fine-grid baseline more than doubles its refinement cost; this configuration is profiled only. Local attention of the same width costs 20.6 GMACs on the adaptive lattice versus 57.0 on the fine grid. The smaller unit count creates room to enrich each unit's processing.

The formulation offers useful accuracy–cost operating points. The five-update main model uses 796.8 GMACs, compared with 732.4 for the released reference. With four updates, it improves all six metrics at 706.1 GMACs. The narrower DACE-64 also costs less and improves five of six metrics, with KITTI EPE favoring the reference. Appendix E gives the full refinement sweep. These results show how compact correspondence processing can support accurate dense flow while allowing a lower-computation operating point.

5 Conclusion

DACE learns the units of correspondence for dense optical flow. A differentiable assignment groups pixels into tokens and maps their motion back to pixels, so grouping and matching are learned jointly. Motion is estimated on coarser tokens, while image evidence stays dense. DACE achieves the lowest zero-shot Sintel Final and KITTI EPE among published methods, with fewer MACs than most. Even against a fixed grid with four times as many units, DACE lowers EPE on every dataset, is more robust where no match is visible, and is more precise near motion boundaries. DACE also improves every metric over a fixed grid at its own resolution, which isolates the benefit of adaptivity. Learning the units to match complements learning the evidence used to match them.

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Appendix overview. We examine whether the accuracy gains extend across backbones (Appendix A), define the regional evaluations and test their robustness (Appendix B), and provide additional flow and learned-support visualizations (Appendix C). Appendix D details the supplied-motion reconstruction study, Appendix E extends the accuracy–computation comparison, and Appendix F reports the limits of the current results after target-specific fine-tuning.

Common setting. Unless stated otherwise, we use the DINOv3 systems from Section 4.1: DACE, released WAFT, and the fixed-grid WAFT-H/4 control with local attention. Models are trained on FlyingChairs and FlyingThings3D; Things test results are in distribution, while Sintel and KITTI training-split results are zero-shot. Backbone and fine-tuning comparisons use the separately identified settings below.

A Zero-Shot Transfer across Backbones

DACE improves zero-shot EPE with both tested backbones. Table 6 compares DACE and WAFT with DINOv3 and Depth Anything V2 (DAv2) [22]. All models use Chairs–Things flow training; WAFT scores are its published results [1].

DACE lowers EPE on both Sintel renderings and KITTI with either backbone. KITTI Fl also improves with DINOv3 and ties at the reported precision with DAv2. The benefit therefore persists across these two pretrained feature representations.

Table 6: **DACE lowers EPE with both backbones.** Zero-shot results on the Sintel and KITTI training splits after training on FlyingChairs and FlyingThings3D. Bold: better of each pair.

		Sintel		KITTI	
Backbone	Method	Clean	Final	EPE	Fl (%)
DINOv3	WAFT [1]	1.28	2.56	3.49	12.9
	DACE	**1.03**	**2.10**	**3.11**	**11.9**
DAv2	WAFT [1]	1.01	2.49	3.28	**10.9**
	DACE	**0.97**	**2.30**	**2.93**	**10.9**

B Regional Evaluation and Robustness

The regional analyses test motion estimation without direct target observations and the separation of neighboring motions. We define the pixel sets used in Section 4.3, explain their contribution to overall error, and examine additional boundary thresholds and large displacements. Regional EPE pools errors over the valid pixels in each specified set; boundary regions can overlap either visibility category.

B.1 Occluded and out-of-frame pixels

Pixel sets. A pixel is out of frame if its ground-truth endpoint $x + F^*(x)$ falls outside Image 2. On Sintel and KITTI, we combine these pixels with those whose targets are marked occluded by the dataset annotations. On Things, we use endpoint bounds only, so this set contains out-of-frame pixels. All sets are restricted to the valid pixels used for standard EPE.

Sintel's official invalid mask further excludes pixels from both visibility categories. These pixels can remain in the standard-EPE valid set, so the two categories need not exhaust that set. Pixel coverage always uses the full standard-EPE valid set as its denominator.

Contribution to overall error. Occluded or out-of-frame pixels occupy 6.4–13.9% of valid pixels but account for 42.3–52.2% of WAFT's standard EPE (Table 7). This concentration shows why the overall score alone does not explain where substantial errors remain, even with fine-grid correspondence.

Table 7: **A small pixel subset accounts for a large share of WAFT's error.** Occluded/out-of-frame pixel coverage and contribution to standard EPE for the released DINOv3 WAFT reference. Coverage pools valid pixels; error contribution retains standard EPE's equal weighting of images. Things uses out-of-frame pixels only.

	Things		Sintel		KITTI
	Clean	Final	Clean	Final	
Pixel coverage (%)	6.4	6.4	8.4	8.4	13.9
Contribution to EPE (%)	43.0	42.7	52.2	42.3	51.9

To measure this contribution, we retain the image weighting used by standard EPE. For K image pairs, let V_i be the standard-EPE valid set of pair i, $O_i \subseteq V_i$ its occluded/out-of-frame set, and $e_i(x) = \|\widehat{F}_i(x) - F_i^*(x)\|_2$ the endpoint error. If E is the standard EPE over these pairs, the fraction contributed by this set is

$$B_O = \frac{1}{KE} \sum_{i=1}^{K} \frac{\sum_{x \in O_i} e_i(x)}{|V_i|}. \tag{8}$$

Each inner term is the region's contribution to that image's mean error; averaging these contributions and dividing by E gives the reported share. The numerator and denominator use the same evaluation pass.

B.2 Accuracy near motion boundaries

Defining the evaluation region. To test accuracy where neighboring motions differ, we measure EPE over a region we define around ground-truth motion boundaries. We identify these boundaries from local changes in ground-truth flow F^*, with strength

$$s(x) = \|\Delta_1 F^*(x)\|_2 + \|\Delta_2 F^*(x)\|_2, \tag{9}$$

where each Δ takes the difference to the next pixel along one image axis. A difference contributes only when both pixels have valid labels. For threshold τ, pixels with $s(x) > \tau$ seed the boundary region, which contains valid pixels in a 5×5 window around any seed. We pool endpoint errors over this region. We evaluate $\tau = 2, 3, 4$ pixels; the main paper uses $\tau = 2$.

Boundary precision improves across the tested thresholds. DACE has lower EPE in these neighborhoods than both fixed-grid references on every dataset and rendering at all three thresholds (Table 8). Released WAFT uses four times as many correspondence units, while WAFT-H/4 with local attention uses the same number as DACE. Thus, the observed boundary advantage persists when the threshold changes, against both finer indexing and a coarse grid with local context.

Table 8: **DACE has lower EPE near motion boundaries at all tested thresholds.** EPE is pooled over a region we define around ground-truth motion boundaries: valid pixels in 5×5 windows around boundary seeds. The threshold sets how large a local flow change must be to count as a boundary. WAFT estimates motion for four times as many units as DACE, and WAFT-H/4 with attention for the same number.

	Things		Sintel		KITTI
System	Clean	Final	Clean	Final	EPE
(a) Boundary threshold: 2 pixels					
WAFT	13.40	13.14	7.30	11.92	9.97
WAFT-H/4 + attention	10.77	10.65	6.97	11.40	11.75
DACE	**9.24**	**9.48**	**6.63**	**10.45**	**8.51**
(b) Boundary threshold: 3 pixels					
WAFT	12.50	12.37	7.38	12.52	12.84
WAFT-H/4 + attention	10.08	10.07	7.02	11.90	15.30
DACE	**8.56**	**8.77**	**6.64**	**11.03**	**10.65**
(c) Boundary threshold: 4 pixels					
WAFT	12.29	12.26	7.75	13.38	15.75
WAFT-H/4 + attention	10.02	9.99	7.34	12.73	18.87
DACE	**8.40**	**8.63**	**6.85**	**11.78**	**12.76**

B.3 Missing targets with large displacements

We next test whether the missing-target improvement persists for pixels with large motion. Following the 40-pixel cutoff in Sintel's $s40+$ category [33], we intersect each occluded/out-of-frame set with pixels whose ground-truth flow magnitude exceeds 40 pixels, using the same cutoff on all datasets.

DACE lowers EPE relative to both references on every evaluated dataset and rendering (Table 9). The improvement therefore remains when absent target observations coincide with large displacement.

Table 9: **DACE improves occluded/out-of-frame flow even at large displacements.** EPE pooled over pixels whose target is occluded or out of frame and whose ground-truth displacement exceeds 40 pixels. Pixel share: this subset's percentage of valid pixels.

	Things		Sintel		KITTI
System	Clean	Final	Clean	Final	EPE
WAFT	33.03	31.41	19.19	29.00	15.34
WAFT-H/4 + attention	27.19	25.25	17.25	27.48	15.81
DACE	**23.77**	**22.73**	**16.16**	**24.08**	**12.36**
Pixel share (%)	4.4	4.4	2.7	2.7	12.2

C Qualitative Flow and Learned Support

These examples connect flow comparisons with learned assignments in the same image regions, using the zero-shot DINOv3 models from Figure 5. Figure 6 adds assignment details for the main paper's departing vehicle. Figure 7 adds two comparisons with clear improvements over WAFT, spanning crossing structures and a broad moving surface.

Reading the comparisons. Each example uses one contextual crop with the same coordinates in both input frames, the visibility map, ground truth and both predictions. All panels share a 2:1 aspect ratio. The input pair establishes movement; the aligned flow maps show the resulting motion. These selected examples illustrate the improvements; the quantitative tables assess performance across the datasets.

Reading the assignments. The box in Image 1 locates the enlarged image, learned-support and assignment-entropy details. White contours separate pixels whose strongest assignments select different units; the model itself uses soft weights. Assignment entropy, normalized by $\log 9$, ranges from dark blue (weight concentrated on one unit) to yellow (weight spread across units). It describes assignment spread, not flow confidence. These maps let the reader inspect the grouping; accuracy is assessed from flow comparisons and the quantitative results.

Visualization protocol. Ground truth and both predictions share the Middlebury color wheel within each pair. On Sintel, magnitude m uses radius $\log(1 + m)/6$; on KITTI, radius grows linearly up to the largest valid ground-truth magnitude. Black in ground-truth flow denotes missing labels. Sparse KITTI labels are enlarged only for display. In target-visibility maps, blue denotes occluded/out-of-frame pixels, gray other valid pixels, and black excluded pixels.

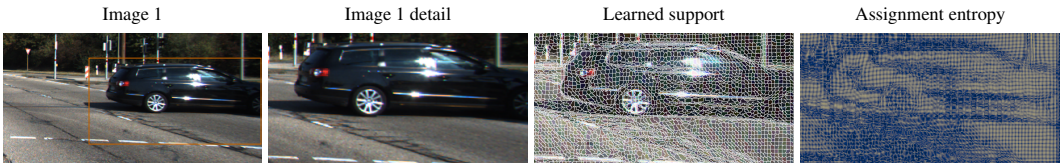

Figure 6: **Learned support for the departing vehicle.** The vehicle is the same as in Figure 5(c). The box in Image 1 locates the enlarged image, learned-support and assignment-entropy panels. Its flow comparison is shown only in the main paper.

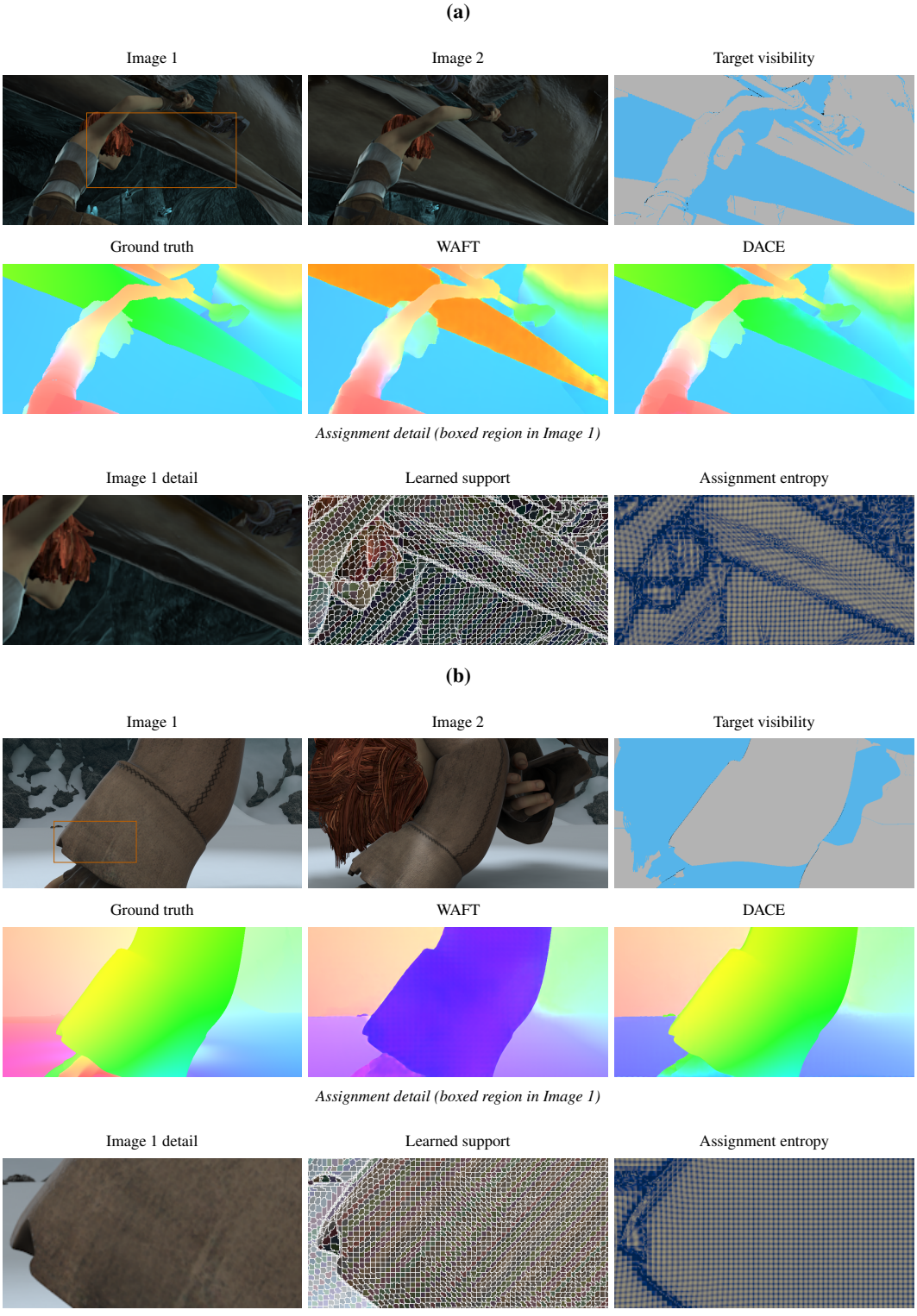

Figure 7: **DACE recovers motion on both thin and broad structures.** (a) WAFT assigns incorrect motion to the wing behind the crossing limbs; DACE recovers the wing's motion. (b) DACE recovers the sleeve's spatially varying motion. Both are Sintel Clean pairs. Marked details show the learned support and assignment entropy.

D Supplied-Motion Reconstruction Protocol

This diagnostic tests how faithfully the units and their readout retain dense motion when predicted unit flow is replaced with pooled ground truth. The results appear in Table 3(c) of Section 4.4.

Substituting pooled ground-truth flow. We first obtain each model's assignments and readout weights from the input images. For DACE, we pool ground-truth flow into the units and broadcast it back through the same assignments, using Eq. (1). For WAFT-H/4 with local attention, we average ground-truth flow within each fixed cell and upsample it with the predicted convex weights. These image-predicted weights remain fixed during the substitution; neither model is retrained. Pooling and evaluation use valid ground-truth pixels. Reconstruction EPE is averaged within each image and then over images.

What the reconstruction measures. Lower error means that fewer motion differences are lost through the combined support and readout. DACE's lower reconstruction error across the evaluated datasets supports its ability to retain dense-motion detail at the same $H/4$ unit count. The comparison evaluates support and readout together; it does not isolate their effects or decompose the end-to-end prediction error.

E Accuracy and Computation across Refinement Steps

Reducing refinement steps provides a lower-computation operating point without retraining. Table 10 evaluates the same trained DACE and WAFT checkpoints with one to five updates. Both were trained with five updates, and their weights remain fixed during this sweep. The separately trained DACE-64 variant halves the recurrent-state width and uses five updates.

Computation is measured for one 540×960 image pair with model-specific padding. Counts include feature extraction, unit matching-feature construction, all refinement updates, and dense-flow readout.

Fewer updates retain a favorable accuracy–computation trade-off. Four-update DACE improves all six metrics over five-update WAFT at 706.1 versus 732.4 GMACs. With three updates, it improves five of six metrics, with KITTI Fl favoring WAFT, at 16% lower computation. DACE-64 also improves five of six metrics, with KITTI EPE favoring WAFT, at 13% lower computation.

At each matched update count, DACE improves all six metrics, while using more computation. The five-update models cost 796.8 and 732.4 GMACs, respectively. Fewer updates let DACE retain the accuracy advantage at a lower-computation operating point.

Table 10: **Four-update DACE outperforms five-update WAFT at lower cost.** Both models are trained with five updates and run here with one to five. DACE-64 halves the recurrent state width.

		Things		Sintel		KITTI		
System	Updates	Clean	Final	Clean	Final	EPE	Fl (%)	MACs (G)
WAFT	1	6.70	6.44	2.20	3.50	8.47	37.1	391.8
DACE	1	5.01	4.84	1.54	2.67	8.30	36.9	433.8
WAFT	2	4.27	4.15	1.49	2.88	4.67	18.5	477.0
DACE	2	2.91	2.84	1.10	2.29	4.09	17.1	524.5
WAFT	3	3.78	3.66	1.34	2.70	3.81	14.0	562.1
DACE	3	2.55	2.49	1.06	2.22	3.38	13.2	615.3
WAFT	4	3.61	3.51	1.29	2.56	3.56	13.1	647.3
DACE	4	2.44	2.43	1.03	2.15	3.18	12.2	706.1
WAFT	5	3.54	3.44	1.28	2.56	3.49	12.9	732.4
DACE	5	2.39	2.38	1.03	2.10	3.11	11.9	796.8
DACE-64	5	2.55	2.48	1.03	2.24	3.52	12.5	636.3

F Target-Specific Fine-Tuning

The zero-shot advantage does not carry over to the current fine-tuned models. Table 11 compares DACE, fine-tuned with WAFT's recipe and submitted to the benchmark servers, with WAFT's published DINOv3 results [1]. These results use the hidden test sets of Sintel, KITTI 2015, and Spring, with target-benchmark training rather than the Chairs–Things-only setting above.

DACE matches WAFT's reported Sintel Final EPE and trails it on Sintel Clean, both KITTI metrics, and all four Spring metrics. Extending the observed transfer gains to target-specific fine-tuning remains an open direction.

Benchmark metrics. Sintel reports EPE. KITTI reports Fl over all labeled pixels (Fl-all) and non-occluded pixels (Fl-noc). Spring provides four ground-truth values per image pixel and measures error against the closest one. Its 1px metric is the percentage of pixels with endpoint error above one pixel, and WAUC summarizes accuracy across error thresholds. Lower is better except for WAUC.

Table 11: **After fine-tuning, DACE matches WAFT on Sintel Final but trails it elsewhere.** Benchmark test results for DACE and DINOv3 WAFT. Lower is better except WAUC (↑).

	Sintel		KITTI 2015		Spring			
System	Clean	Final	Fl-all (%)	Fl-noc (%)	1px (%)	EPE	Fl (%)	WAUC ↑
WAFT	0.94	2.02	3.56	2.13	3.18	0.33	1.25	95.05
DACE	1.06	2.02	3.67	2.35	3.43	0.35	1.30	94.86